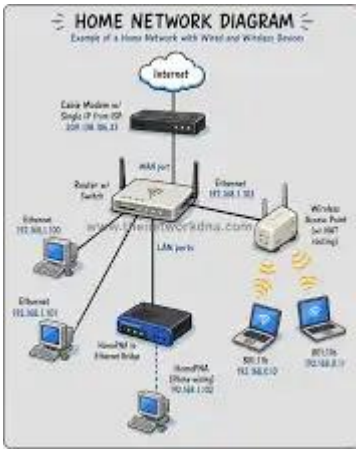
Network devices

Q-1 Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

Ans:



Explanation

1. Internet: Provides the internet connection to the home.
2. Wi-Fi Router: Connects to the internet and provides wireless connectivity.
3. Network Switch: Connects multiple wired devices to the local network.
4. Laptop: Connects to the switch using an Ethernet cable.
5. Smartphone: Connects wirelessly to the router through Wi-Fi.
6. Smart TV: Connects to the switch using an Ethernet cable for reliable video streaming.

Q-2 Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Ans: Comparison of Network Devices and Their Real-World Uses

Network Device	**Main Function**	**Real-World Use Case**
Router	Connects different networks and provides internet access.	Used at home to connect smartphones and laptops to the internet for watching YouTube videos.
Switch	Connects multiple devices within a local area network (LAN).	Used in a college computer lab to connect computers for file sharing and LAN gaming.

Hub	Connects multiple devices and broadcasts incoming data to all connected ports.	Can be used in a small practice network to demonstrate basic data communication.
Access Point (AP)	Provides Wi-Fi connectivity to wireless devices.	Used in a college to provide Wi-Fi access for students using WhatsApp, Google Classroom, and YouTube.

Q-3 Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Ans: Scenario: Ordering Food on Zomato

When I order food using the Zomato app on my smartphone, data travels through several network devices to reach the internet.

1. Access Point (AP): My smartphone connects to the home Wi-Fi network through the access point. It sends my food order data wirelessly.

2. Router: The router receives the data and forwards it towards the internet. It helps connect my home network to the internet.

3. Switch: If a switch is used in the home or network infrastructure, it forwards data between connected wired devices, such as the router and other network equipment.

4. Internet and Zomato Server: The data reaches Zomato's servers, where my order request is processed. The restaurant receives the order details.

5. Response to My Phone: Confirmation of my order travels back through the internet and router to the access point, which sends it wirelessly to my smartphone.

Q-4 Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

Ans: Two Ways to Connect a Gaming PC to the Internet

If my gaming PC is far from the Wi-Fi router and has only an Ethernet port, I can use these two network devices.

1. Powerline Adapter

- How it works: One powerline adapter connects to the router using an Ethernet cable and plugs into a wall socket. The second adapter plugs into a wall socket near the gaming PC and connects to the PC using another Ethernet cable.
- Benefit: It uses the home's electrical wiring to carry network data, so I may not need a long Ethernet cable.
- Use case: Suitable for online gaming when the router is in another room.

2. Wi-Fi Extender with an Ethernet Port

- How it works: The Wi-Fi extender receives the router's Wi-Fi signal and extends coverage. I connect the gaming PC to the extender's Ethernet port using an Ethernet cable.
- Benefit: It allows a PC without Wi-Fi to access the network wirelessly through the extender.
- Use case: Useful when the PC is far from the router but the extender can receive a good Wi-Fi signal.

Comparison

Feature	**Powerline Adapter**	**Wi-Fi Extender**
Connection method	Home electrical wiring	Wi-Fi signal
PC connection	Ethernet cable	Ethernet cable
Main advantage	Can provide a stable connection	Easy to install without relying on electrical wiring quality
Limitation	Performance depends on electrical wiring	Performance depends on Wi-Fi signal strength and interference